

Hands on metabarcoding - Shared microbiome

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Shared microbiome (1)

If we want to know what taxa are shared between the microbiome after different treatments, we can use the classical venn diagram approach.

To plot a venn diagram, we can use the R package `ggVennDiagram`, which produces venn diagrams using `ggplot` grammar.

Shared microbiome (2)

Let's start with the basic example of shared communities between the control and the treatment with high concentrations of Cd.

```
# create a vector with ASVs from each treatment
ctr_asvs <- ASVs_full %>% filter(Treatment == "CTR") %>% pull(Sequence)
#
cd_15_asvs <- ASVs_full %>% filter(Treatment == "____") %>% pull(Sequence)
```

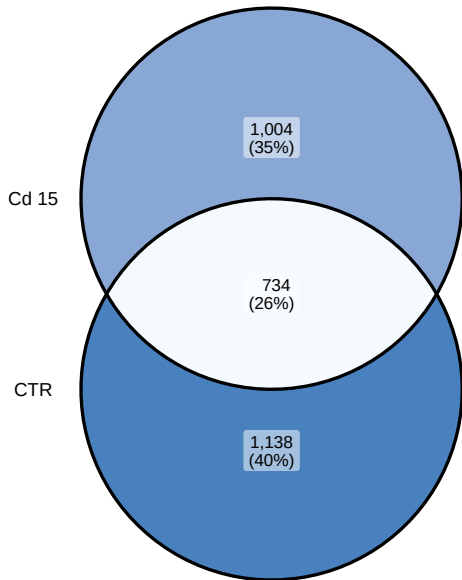
Shared microbiome (3)

```
# create a vector with ASVs from each treatment
ctr_asvs <- ASVs_full %>% filter(Treatment == "CTR") %>% pull(Sequence)
#
cd_15_asvs <- ASVs_full %>% filter(Treatment == "Cd 15") %>% pull(Sequence)
#
```

Shared microbiome (4)

```
ggVennDiagram(x = list(ctr_asvs, # group 1
                        cd_15_asvs), # group 2
              category.names = c("CTR",
                                "Cd 15"),
              label_size = 4) +
  scale_fill_gradient(low = "#F4FAFE", high = "#4981BF") +
  guides(fill = "none") +
  xlim(-6,9) +
  ylim(-5,10)
```

Shared microbiome (5)



Try to do a venn diagram with all treatments (1)

First make a vector with the ASVs from each treatment:

```
# make a vector with the ASVs of each missing treatment  
cd_0.015_asvs <- ASVs_full %>% filter(____ == ____ ) %>% pull(____)  
cd_0.15_asvs <- ASVs_full %>% filter(____ == ____ ) %>% pull(Sequence)  
cd_1.5_asvs <- ASVs_full %>% filter(____ == ____ ) %>% pull(Sequence)
```

Try to do a venn diagram with all treatments (2)

First make a vector with the ASVs from each treatment:

```
# make a vector with the ASVs of each missing treatment
```

```
cd_0.015_asvs <- ASVs_full %>% filter(Treatment == "Cd 0.015") %>% pull(Sequence)
```

```
cd_0.15_asvs <- ASVs_full %>% filter(Treatment == "Cd 0.15") %>% pull(Sequence)
```

```
cd_1.5_asvs <- ASVs_full %>% filter(Treatment == "Cd 1.5") %>% pull(Sequence)
```


Try to do a venn diagram with all treatments (3)

Then you can plot in the same way:

```
# may take a while to plot
ggVennDiagram(x = list(____,
                        _____,
                        _____,
                        _____,
                        _____),
              category.names = c("CTR (0 \U03BCM)",
                                "Cd 0.015 \U03BCM",
                                "Cd 0.15 \U03BCM",
                                "Cd 1.5 \U03BCM",
                                "Cd 15 \U03BCM"),
              label_size = 4) +
guides(fill = "none")
```

Try to do a venn diagram with all treatments (4)

Then you can plot in the same way:

```
# may take some time to plot
ggVennDiagram(x = list(ctr_asvs,
                        cd_0.015_asvs,
                        cd_0.15_asvs,
                        cd_1.5_asvs,
                        cd_15_asvs),
              category.names = c("CTR (0 \U03BCM)",
                                "Cd 0.015 \U03BCM",
                                "Cd 0.15 \U03BCM",
                                "Cd 1.5 \U03BCM",
                                "Cd 15 \U03BCM"),
              label_size = 4) +
#scale_fill_gradient(low = "#F4FAFE", high = "#4981BF") +
guides(fill = "none")
```

Try to do a venn diagram with all treatments (5)

